

Nehal Choudhary

(720) 377-5333 | nechoudhary@ucsd.edu | github.com/ncdev06 | nehal-choudhary-7a3516202/

EDUCATION

University of California San Diego

June 2027

B.S. Mathematics-Computer Science, B.S. Cognitive Science-Machine Learning

3.9

- Relevant Coursework: Data Science in Practice, Modeling & Data Analysis, Probability, Linear Algebra, Econometrics, Advanced Data Structures, Discrete Math & Graph Theory, Systems Programming & Software Tools

EXPERIENCE

Machine Learning Intern | Netcracker Technology

July 2025 - September 2025

- Built an **LLM evaluation framework** testing **120+ scenarios across 6 reliability metrics**, replacing manual prompt testing with structured benchmarking.
- Ran **prompt A/B experiments across 20+ variants** to diagnose hallucination, scope drift, and schema instability, improving model response consistency.
- Diagnosed **state management and retrieval transparency failures** in production workflows, proposing architectural fixes to improve contextual continuity and grounding.

Machine Learning Researcher | Cognitive Development Lab @ UCSD

December 2025 - Present

- Refined a **multimodal evaluation pipeline** processing **300+ real-world videos**, ensuring reproducible experimentation across heterogeneous environments.
- Conducted the lab's **first ASR benchmarking comparisons (Whisper vs. human labels)** across **300+ segments**, measuring similarity ($\tilde{0.75}$) and transcription accuracy ($\tilde{63\%}$).
- Performed robustness analysis of **YOLOv5 detection across 4+ environments** (up to **0.99** precision), surfacing failure regimes and refining annotations to strengthen evaluation validity.

PROJECTS

Neural Time-Series Forecasting Hackathon, 2nd Place *PyTorch · GRU · LSTM · Temporal Models*

- Built **deep sequence models (GRU, LSTM, TCN)** for multivariate neural signal forecasting across **80–240 channels with 10-step predictions**.
- Designed an **LSTM forecasting architecture** using delta-based supervision, teacher forcing, and normalization, reducing MSE from **300k → 50k**.
- Deployed a full **ML training pipeline (data preprocessing, training, checkpointing, evaluation)** enabling rapid experimentation and contributing to a **2nd place finish**.

Spotify Similarity Recommender *Python · FAISS · pandas · Streamlit*

- Engineered a **FAISS-based music similarity engine** over **900+ Spotify tracks**, enabling fast cosine similarity search on audio features.
- Architected an **ETL pipeline validating Spotify Web API feature data**, ensuring consistent schema and reliable similarity comparisons.
- Developed an **interactive Streamlit dashboard** to explore track similarity and embedding relationships through user-driven queries.

ClassGraph, 1st Place *Python · React · Go*

- Created a **course-planning platform modeling 1,000+ UCSD courses as directed graphs**, mapping **100+ prerequisite chains**.
- Implemented **recursive prerequisite validation and graph traversal algorithms** to detect scheduling constraints and degree pathway conflicts.
- Integrated **React + Go backend infrastructure** for interactive prerequisite visualization and scalable academic planning.

SKILLS AND TOOLS

Languages: Python, C++, Java, SQL, JavaScript, Stata, HTML/CSS

ML & AI: PyTorch, scikit-learn, FAISS, Whisper, LLM Evaluation, Time-Series Models (LSTM, GRU, TCN), Object Detection (YOLO)

Data & ML Infrastructure: pandas, NumPy, ETL Pipelines, Model Evaluation, A/B Testing

Tools: Git, Streamlit, CVAT